TECTONICS, VOL. 6, NO. 5, PAGES 573-583, OCTOBER 1987

ORDOVICIAN DOCKING OF THE CAROLINA SLATE BELT: PALEOMAGNETIC DATA

H. K. Vick, J. E. T. Channell, and N. D. Opdyke

Department of Geology, University of Florida, Gainesville

Abstract. Paleomagnetic results from eight sites in argillaceous metasediments from Union County, North Carolina, in the central Carolina slate belt yield a consistent high blocking temperature magnetization component. The precision of the mean declination and inclination of all site means is maximized at 50% tilt correction, and the improvement in grouping relative to 0% and 100% tilt correction is significant at the 95% confidence level. This tends to indicate that the high blocking temperature component was acquired during folding. The metamorphism and deformation in this area are rather well constrained, by recent $^{40}Ar/^{39}Ar$ data, to the Late Ordovician (~450 Ma). We therefore interpret the computed pole position (29.6°N, 122.1°E, α_{95}=5.11) as Late Ordovician in age and note that it lies close to some of the Late Ordovician poles for cratonic North America. The paleolatitude (22.2°S) of the sampling site is certainly unlike those derived from Armorica and Gondwana for this time interval and is rather similar to that proposed for the Iapetus coast of Laurentia during the Late Ordovician. Indeed, the magnetization (and deformation) may be contemporaneous with the docking of the Carolina slate belt onto the Laurentian continent.

INTRODUCTION

The Carolina slate belt extends for some 640 km from Virginia to central Georgia (Figure 1). It comprises Precambrian to Cambrian volcanic and sedimentary rocks of island arc affinity, metamorphosed to lower greenschist facies. The Carolina slate belt has been loosely associated with the Avalon terrane [Williams and Hatcher, 1982]. Although there is no clear evidence for this association, it does seem probable, on the basis of isolated fossil finds, that the slate belt originated in a faunal province distinct from that of the North American continental margin [Secor et al., 1983].

Paleomagnetic data have provided useful constraints on early Paleozoic paleogeography although there is, at present, no clear consensus. It has been suggested that Avalon, as well as Armorica and Gondwana, was at high southerly latitudes relative to Laurentia during the early Paleozoic [Van der Voo and Johnson, 1985; Perroud et al., 1984]. Whereas Avalon and Armorica may have reached their present position relative to Laurentia in the middle Paleozoic, the formation of Pangea culminated in the Hercynian collision of Gondwana in the late Paleozoic. Early Paleozoic paleomagnetic data from the Carolina slate belt should indicate whether this island arc was part of Laurentia or Avalon, or was an independent terrane during this time.

Throughout the Carolina slate belt, the primary Precambrian to Cambrian magnetizations have been destroyed by metamorphism. However, the $^{40}Ar/^{39}Ar$ technique of dating thermotectonic events in low-grade metamorphic rocks provides a potential means of dating magnetization components acquired during metamorphism. This study and that of Noel et al. [1987] are first efforts at correlating high blocking temperature magnetization components to $^{40}Ar/^{39}Ar$ plateau ages in Carolina slate belt rocks.

GEOLOGIC SETTING

The sampling sites for this study are located in Union County, North Carolina, within the central Carolina slate

Copyright 1987
by the American Geophysical Union.

Paper number 7T0463.
0278-7407/87/007T-0463$10.00

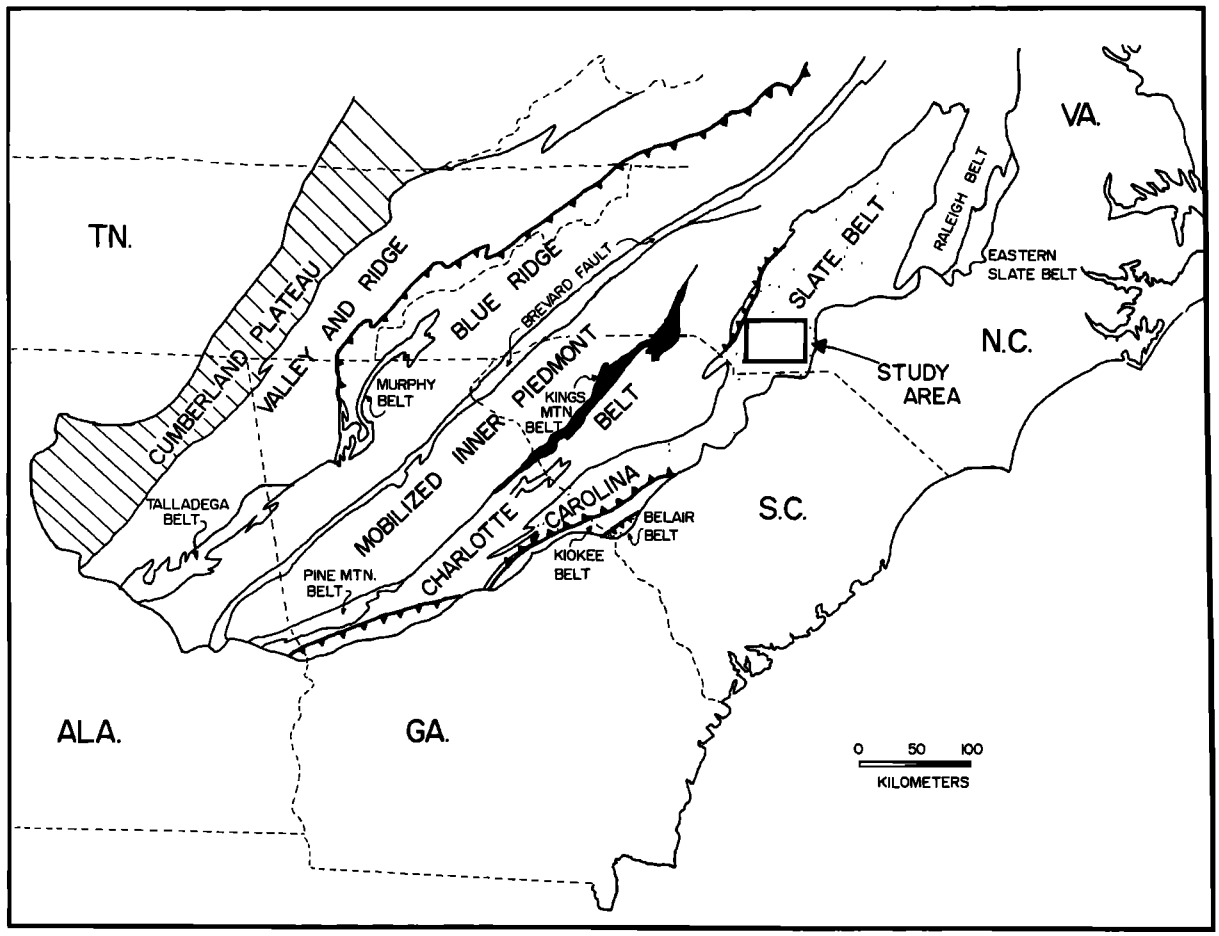


Fig. 1. The Carolina slate belt in relation to other regional tectonic belts in the southern Appalachians [after Hatcher, 1972; Hatcher and Zietz, 1980].

belt (Figure 1). The area has been mapped by Randazzo [1972] and Milton [1984] (Figure 2). Oriented drill core samples were collected in the volcaniclastic mudstones of the middle Cid Formation of the Albemarle Group. The Albemarle Group rocks are underlain by the Uwharrie Formation, a thick felsic pyroclastic deposit which is the oldest locally exposed unit. The Albemarle Group comprises, from bottom to top, the Tillery, Cid, Floyd Church, and Yadkin formations. Each of these four units is a volcaniclastic sequence, with volcanic and tuffaceous intervals [Milton, 1984]. The Cid Formation and the conformably underlying Tillery Formation are argillaceous volcaniclastic mudstones characterized by thin beds, 1-40 cm in thickness. The presence of Ediacarian fauna in float thought to have come from the Floyd Church Formation suggests a Late Precambrian age for this part of the sequence [Gibson et al., 1984]. This is consistent with U-Pb isotopic data on zircons from a tuff in the Uwharrie Formation which have been interpreted to give a 586-Ma crystallization age [Wright and Seiders, 1980].

The low-grade metamorphism in these rocks is generally assigned to the muscovite-chlorite and biotite subfacies of the greenschist facies [Stromquist and Sundelius, 1969]. In this part of the central slate belt, bedding is generally well preserved with a well-defined spaced cleavage approximately axial planar to a system of gentle folds trending NE-SW (Figure 2). Kish et al. [1979] reported a whole-rock K-Ar age of 483 ± 15 Ma for slate from the Albemarle Group, which they interpreted as metamorphic cooling age. This is broadly consistent with new $^{40}Ar/^{39}Ar$ plateau ages of 455 ± 2 Ma from the Tillery Formation [Noel et al., 1987] which suggest a Late Ordovician metamorphism in this part of the slate belt. The cleavage represents the metamorphic fabric, and as the cleavage appears to be axial planar, it is reasonable to assume that folding is contemporaneous with metamorphism, and hence both the folding and metamorphism can be considered as Late Ordovician in age. This is consistent with the conclusions of Glover et al. [1983], but it is important to note that considerably younger Devonian metamorphic ages

19449194, 1987, 5, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/TC006i005p00573 by Curtin University Library, Wiley Online Library on [25/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

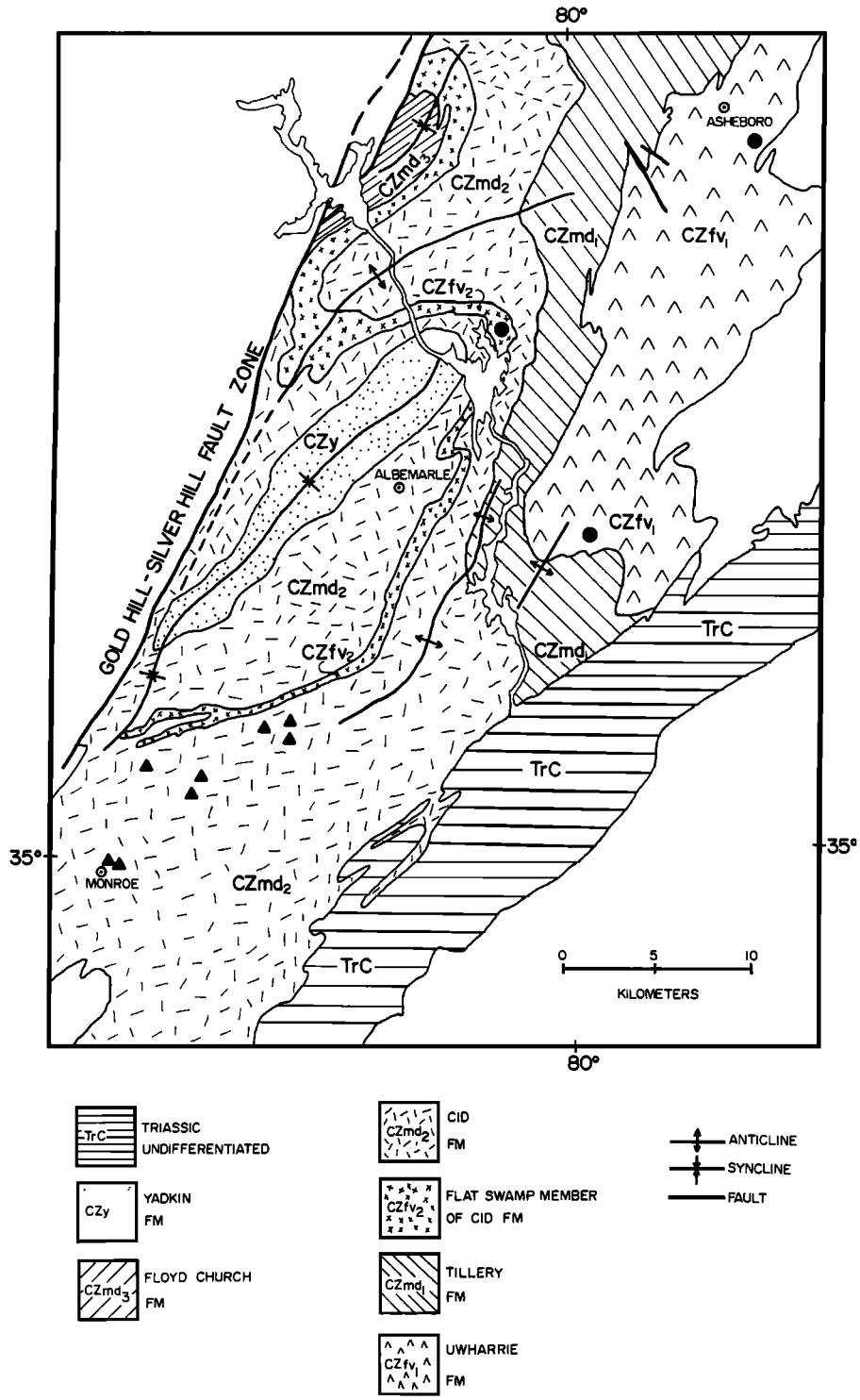


Fig. 2. Geologic map showing major rock units outcropping in the Carolina slate belt between the North Carolina-South Carolina state line and Asheboro, North Carolina. Sampling locations for this study are depicted with solid triangles, while sampling locations from Noel, et al. [1987] are depicted with solid circles [after North Carolina Geological Survey, 1985].

19449194, 1987, 5, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/TC006i005p00573 by Curtin University Library, Wiley Online Library on [25/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

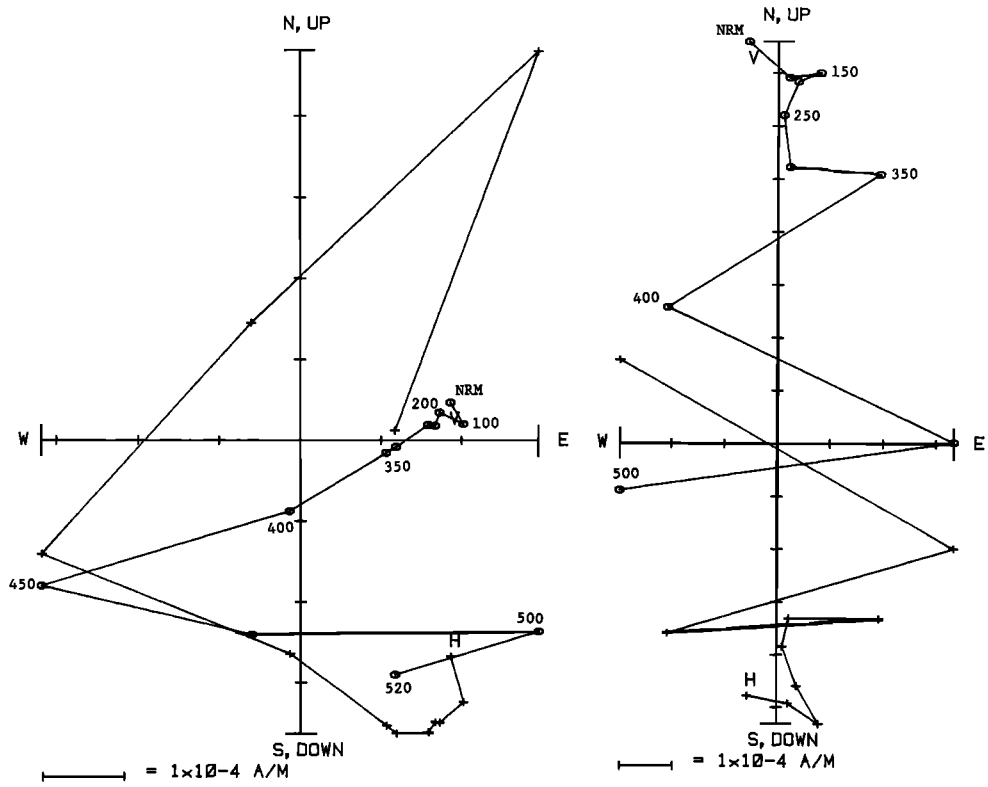


Fig. 3. Orthogonal projections of thermal demagnetization data exhibiting a behavior characteristic of some samples from which a high blocking temperature component could not be isolated. Circles and crosses represent projection on the vertical and horizontal planes respectively. The demagnetization temperatures for some vector end points are indicated in degrees Celsius.

are recorded to the south, near Columbia, South Carolina [Dallmeyer et al., 1986], and that the metamorphic history of the Carolina slate belt is nonuniform.

PALEOMAGNETISM

Of the 26 sampling sites in the Cid Formation, only eight yielded internally consistent data. The rejected sites fell into two categories. For 13 sites, high blocking temperature components defined from the thermal demagnetization data using principal component analysis [Kirschvink, 1980] were associated with unacceptably high mean angular deviation (MAD) values (> 20°). In general, after heating above 350°C, the magnetization of these samples became viscous, and directions became inconsistent (Figure 3). We associate this behavior with oxidation of iron sulphides, to magnetite and other oxides, during the heating cycle. An increase in susceptibility at about 400°C coincides with the onset of this erratic behavior. For the other five rejected sites, high blocking temperature components were associated with MAD values [Kirschvink, 1980] of less than 20°, but the within-site scatter resulted in site means with α_{95} values [Fisher, 1953] greater than 25°.

We restrict our discussion to the eight sites which yielded internally consistent high blocking temperature components. The natural remanent magnetization (NRM) had an average intensity of 5 X 10^{-4} A/m, and the directions, though scattered, were mainly southward with positive inclinations. All samples were progressively thermally demagnetized at an average of 12 steps up to a maximum temperature of 520°-580°C, at which point sample intensities decreased to magnetometer noise level or became too viscous and unstable for precise measurement. Most of the samples contained at least two magnetization components (Figure 4). The low blocking temperature components are generally scattered, but the high blocking temperature components are consistently directed southeastward with positive inclinations. The high blocking temperature components are usually isolated in the 350°-560°C range, and the maximum blocking temperatures do not exceed 570°C.

Alternating field demagnetization is also often effective at isolating this component (Figure 5). The relatively low coercivities indicated by the alternating field treatment and by the acquisition curves of isothermal remanence (IRM) (Figure 6), together with the maximum blocking temperature, point to magnetite as the principal remanence carrier in these rocks. We suggest that the formation of

19449194, 1987, 5, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/TC006i005p00573 by Curtin University Library, Wiley Online Library on [25/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

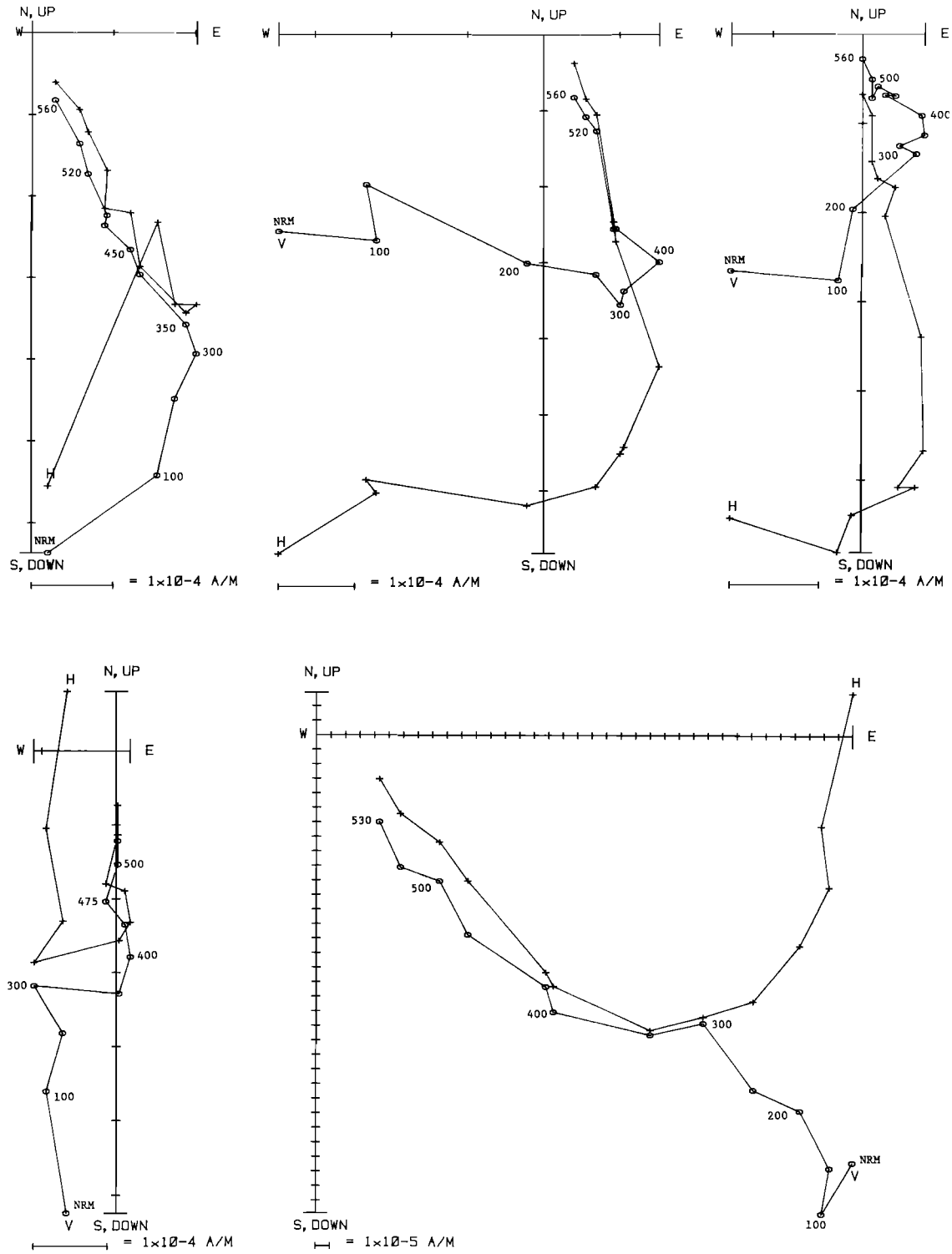


Fig. 4. Orthogonal projections of thermal demagnetization data of representative samples from five of the sites from which a high blocking temperature component could be resolved. Circles (crosses) represent the projection of vector end points on the vertical (horizontal) plane. The demagnetization temperatures are indicated for some vector end points.

19449194, 1987, 5, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/TC006i005p00573 by Curtin University Library, Wiley Online Library on [25/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

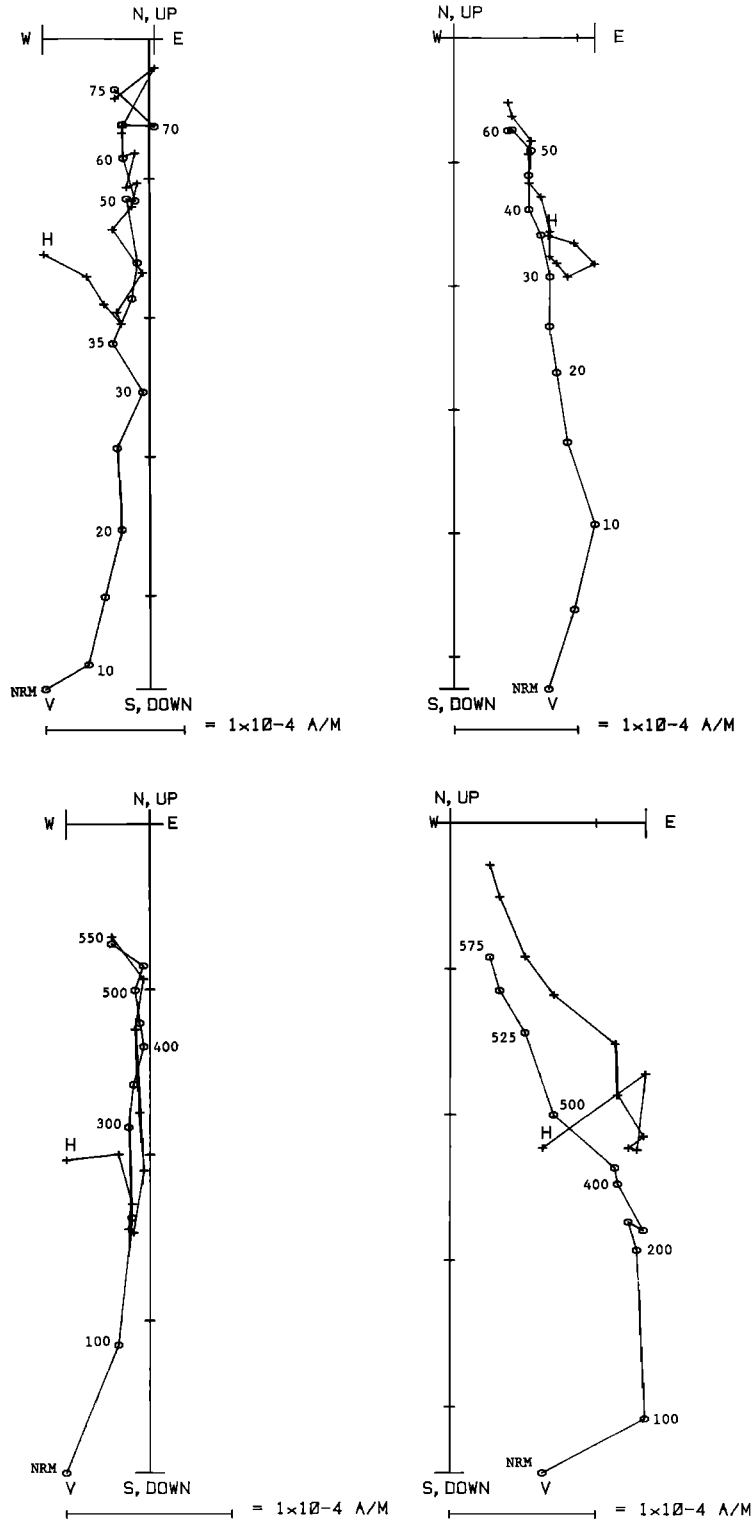


Fig. 5. Orthogonal projection of alternating field demagnetization data (above) compared with projections from thermally demagnetized samples (below) from the same sites. For the alternating field data, the peak fields are shown in milliteslas. For the thermal data the temperatures are indicated in degrees Celsius. Other notation as in Figures 3 and 4.

19449194, 1987, 5, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/TC006i005p00573 by Curtin University Library, Wiley Online Library on [25/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

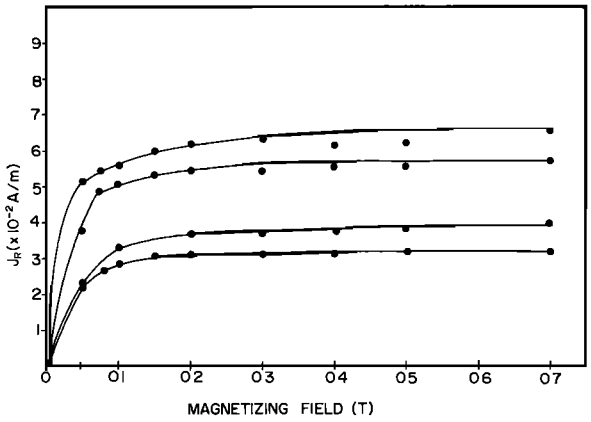


Fig. 6. Acquisition of isothermal remanent magnetization in four samples from different sites, indicating the dominance of low-coercivity minerals.

magnetite during metamorphism, perhaps from more primary iron sulphides, was not ubiquitous but was restricted to particular local conditions.

High blocking temperature magnetization components were computed from straight line segments picked by eye from the orthogonal projections. The principal component analysis advocated by Kirschvink [1980] was used to define the direction, and this direction was considered acceptable if the MAD value was less than 20°. The highest accepted MAD value was 17°, and 80% were less than 10°. The component directions are plotted in Figure 7 and are combined into site mean directions using standard Fisher statistics (Table 1). The grouping of site mean directions before and after structural tilt correction is almost identical (Figure 8, Table 1). However, 50% tilt correction improves the grouping relative to 0% and 100% tilt correction. The improvement in precision (Figure 9) is significant at the 95% confidence level, suggesting that the high blocking temperature component is synfolding. This magnetization component after 50% tilt correction yields a pole position of 29.6°N, 122.1°E.

DISCUSSION

The $^{40}Ar/^{39}Ar$ plateau ages from the Tillery Formation in southern North Carolina indicate a Late Ordovician metamorphic event [Noel et al., 1987]. This is broadly consistent with the conclusions of Kish et al. [1979] and Glover et al. [1983] who suggested an Early Ordovician age for folding and metamorphism in both the Uwharrie Formation and the Albemarle Group. Although Alleghenian and Acadian deformation events are apparent in slate belt rocks further south, there is only evidence for a single metamorphic/deformational event in the southern part of North Carolina [Kish et al., 1979]. Here the metamorphic fabric comprises a spaced cleavage axial planar to D1 folds, indicating that folding and metamorphism were broadly contemporaneous. This D1 fabric can be traced into the higher-grade metamorphic rocks of the Charlotte belt, where a suite of 415- to 385-Ma plutons clearly postdates the fabric, indicating a pre-Devonian age for D1 [Sutter et al., 1983; Butler and Fullager, 1978]. A similar-aged deformational phase is observed in terranes with clear North American affinities [Williams and Hatcher, 1982] whereas Avalonian terranes in

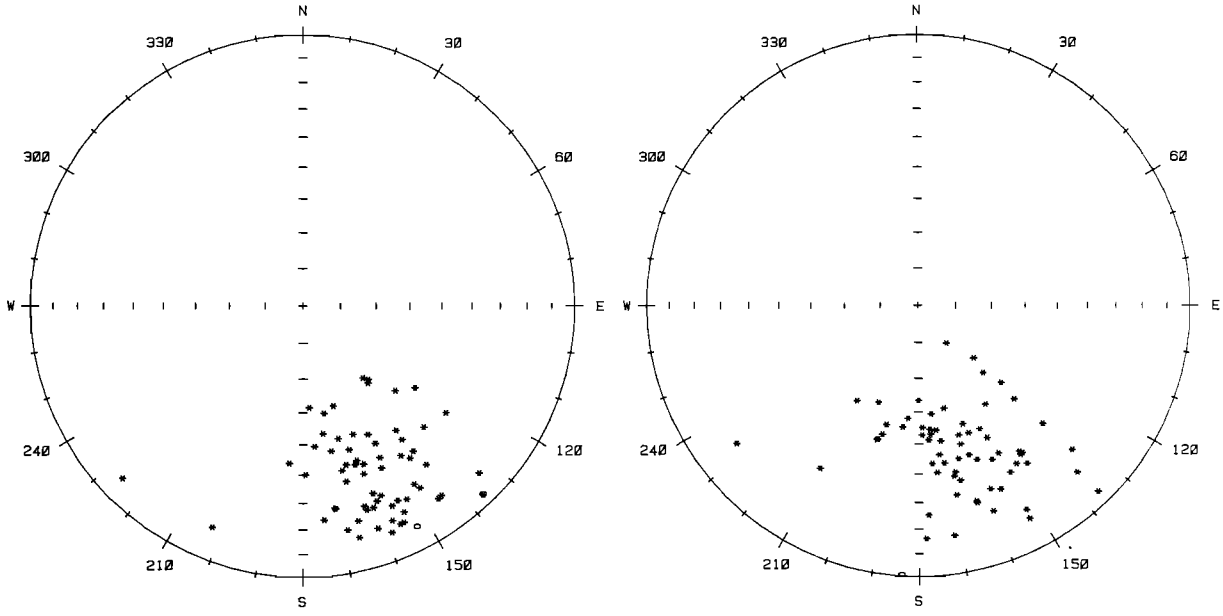


Fig. 7. High blocking temperature component directions plotted in stereo-graphic projection. Stars (circles) represent positive (negative) inclinations. Directions are shown before tilt correction (left) and after 100% tilt correction (right).

19449194, 1987, 5, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/TC006i005p00573 by Curtin University Library, Wiley Online Library on [25/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

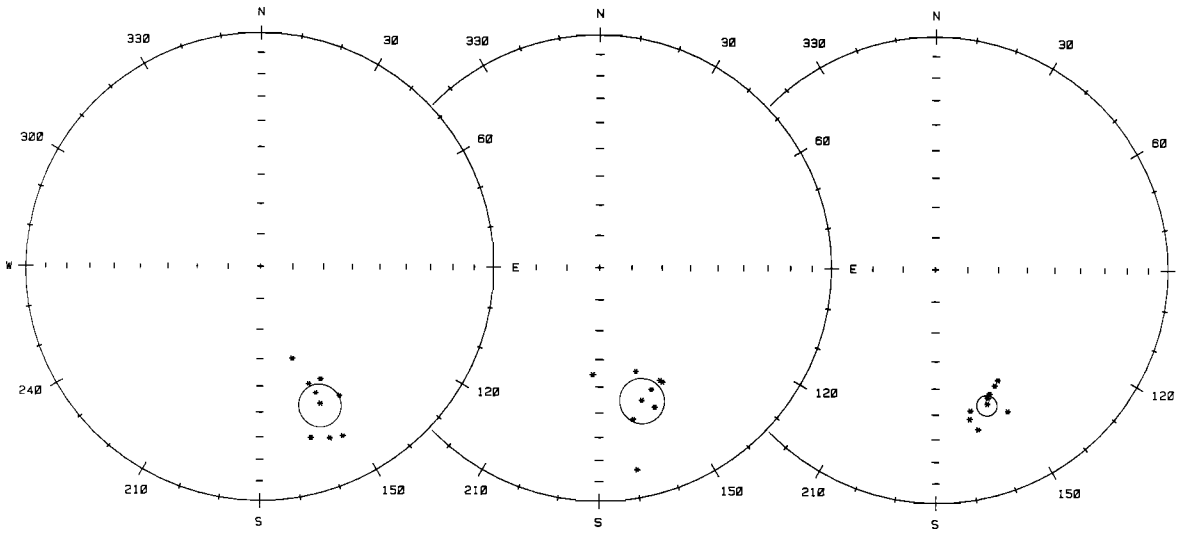


Fig. 8. Site mean directions plotted in stereographic projection. All directions have positive inclination. Directions shown are before tilt correction (left), after 100% tilt correction (center), and after 50% tilt correction (right). The mean direction of each grouping is indicated with a circle of 95% confidence.

Table 1. Mean Declination and Inclination for All Sites Considered in This Study

Site	Declination	Inclination	k	Alpha 95
		Geographic		
11	153.84	21.53	34.7	10.39
12	160.79	56.80	28.2	14.65
13	157.49	45.85	55.2	10.39
15	157.71	22.96	64.3	4.80
16	151.68	45.61	29.0	9.73
18	163.40	25.70	17.6	12.62
27	156.20	41.65	115.6	6.26
28	148.30	36.40	18.8	21.0
Mean	156.13	37.14	37.2	9.17
		50% Tilt Corrected		
11	153.01	33.43		
12	167.20	35.60		
13	157.89	41.30		
15	166.17	38.70		
16	150.60	45.86		
18	165.07	30.83		
27	156.50	42.60		
28	153.03	44.59		
Mean	158.95	39.26	118.54	5.11
		100% Tilt Corrected		
11	151.50	45.30	34.86	10.37
12	169.30	14.06	67.52	9.38
13	158.20	36.90	55.09	10.40
15	183.35	53.13	70.23	4.59
16	151.70	44.34	29.27	9.68
18	167.35	34.89	15.13	13.68
27	156.79	43.55	116.11	6.24
28	160.56	52.13	18.77	21.0
Mean	162.20	41.03	31.06	10.09

19449194, 1987, 5, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/TC006i005p00573 by Curtin University Library, Wiley Online Library on [25/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

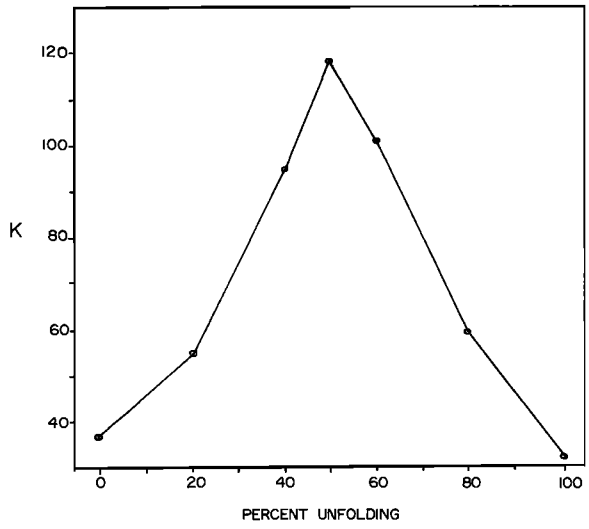


Fig. 9. Precision (k) values [Fisher, 1953] versus percent unfolding for the overall mean derived from eight site means.

Canada and New England do not record this early Paleozoic deformation event [Zen, 1981]. In addition, the stratigraphy tends to indicate that the Carolina slate - Charlotte belt and Avalon are distinct terranes. The implication of the above is that the accretion of the Carolina slate - Charlotte belt to Laurentia is early Paleozoic in age, in contrast to the middle or late Paleozoic accretion age for Avalon [Secor et al., 1986].

The synfolding nature of the high blocking temperature magnetization component allows us to infer an age for this component. The $^{40}Ar/^{39}Ar$ plateau ages imply a cooling of these rocks through the 325°C isotherm at about 455 Ma (Late Ordovician) [Noel et al., 1987]. Peak metamorphic temperatures and cooling rates are poorly constrained, and it is therefore not possible to infer the precise age relationship of the plateau age to the blocking of the magnetization component. However, it seems likely that the component was acquired during the Middle to Late Ordovician.

The question arises as to the effect of strain on the high blocking temperature remanence direction. Even if the orientation and shape of the strain ellipsoid are known, either by use of strain markers or by using magnetic susceptibility anisotropy, there is no satisfactory method of removing the effects of strain from the remanence direction [Lowrie et al. 1986]. It is generally agreed that removal of strain, by inverse strain correction using the passive marker method [Kligfield et al., 1983], will shift the remanence vector toward the Z axis of the strain ellipsoid; however, the determination of the amount of shift remains a problem. In the case of the slate belt sites, the spaced cleavage is subvertical and approximately axial planar to major NE trending folds (Figure 2). Poles to cleavage are grouped with declinations between 140° and 170° and inclinations close to zero. If we assume that the poles to cleavage are close to the Z axis of the strain ellipsoid, the removal of strain would be expected to shallow the remanence inclination but have little effect on the remanence declination.

Noel et al. [1987] have sampled volcanic and volcaniclastic rocks from the Uwharrie and Cid formations. The mean direction of their high blocking temperature component is based on four sites from the Albemarle-Asheboro area, just north of Union County, North Carolina (Figure 2). The magnetization is considered to be postfolding in age, and, although the mean inclination (40°) is close to that which we obtain, the mean declination (220.9°) is very different. Their distribution of component directions is not typically Fisherian, with a larger scatter in declination than in inclination. The more southerly declinations (192.6°, 189.8°), recorded near Albemarle, in the south of their study area and within 30 km of our field area (Figure 2) are closer to the mean declination which we obtain. The variability in declination recorded by Noel et al. [1987], and the discrepancy with our data may be due to local rotations of small-scale tectonic blocks, or to the problem of using poorly resolved components from a small data set.

As mentioned above, any effect of strain in the slate belt rocks would be to steepen the remanence inclination, and therefore the paleolatitude should be considered a

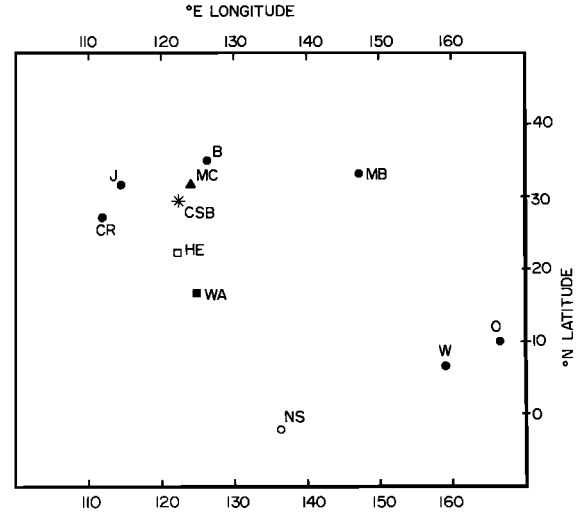


Fig. 10. Plot of paleomagnetic pole positions from cratonic North America and Avalon during the Paleozoic. Solid circles represent the more reliable Laurentian poles from the Lower Ordovician [Jackson and Van der Voo, 1985, [O]; Van der Voo et al., 1976 [W]] and Upper Ordovician [Proko and Hargraves, 1973 [B]; Van der Voo and French, 1977 [J]; Watts and Van der Voo, 1979 [CR, MB]]. The open circle represents the Lower Ordovician pole from Avalon [Van der Voo and Johnson, 1985 [NS]]. A Silurian pole from Laurentia is denoted by a solid square [McCabe et al., 1985 [WA]] and the Siluro-Devonian pole from Avalon by an open square [Kent and Opdyke, 1980 [HE]]. The solid triangle denotes the Mississipian Mauch Chunk pole attributed to Laurentia [Kent and Opdyke, 1985 [MC]]. The star indicates the pole obtained by this study [CSB].

19449194, 1987, 5, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/TC006i005p00573 by Curtin University Library, Wiley Online Library on [25/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

maximum value. The Ordovician paleolatitude inferred from our data set (22°S ± 1) and that of Noel et al. [1987] (22°S ± 4) are consistent with paleolatitudes expected for the Iapetus seaboard of Laurentia in the Ordovician [Perroud et al., 1984]. The paleolatitudes are considerably lower than "Avalonian" Ordovician paleolatitudes from Nova Scotia [Van der Voo and Johnson, 1985], and Delaware [Rao and Van der Voo, 1980]. Hence we conclude that the slate belt was situated at or close to the Laurentian margin of the Ordovician Iapetus ocean, and probably a considerable distance from both the Armorican and Gondwana continents, which were probably situated at southerly latitudes greater than 50° at this time.

The early Paleozoic apparent polar wander path for North America is still poorly defined and is undergoing revision as the extent of the late Paleozoic remagnetizations is realized [e.g. Kent and Opdyke, 1985]. The "reliable" Late Cambrian and Early Ordovician poles [Van der Voo et al., 1976; Jackson and Van der Voo, 1985] are separated from the Late Ordovician poles [Van der Voo and French, 1977; Watts and Van der Voo, 1979] which plot close to the Silurian pole given by McCabe et al. [1985] and to the Early Carboniferous pole from the Mauch Chunk Formation [Kent and Opdyke, 1985] (Figure 10). There is some debate as to whether these Laurentian Late Ordovician poles were remagnetized in Silurian to Early Carboniferous time. However, the pole from the Carolina slate belt clearly plots close to these Late Ordovician poles, which is consistent with close association and possible accretion of the slate belt to Laurentia by the Late Ordovician.

Acknowledgements. Jack Hillhouse and Donald Secor gave us the benefit of thoughtful reviews. We appreciate helpful discussions with Dave Dallmeyer and Dan Sparlosu, who also provided us with a preprint of Noel et al. [1987].

REFERENCES

Butler, J. R. and P. D. Fullager, Petrochemical and geochronological studies of plutonic rocks in the southern Appalachians, III, Leuco-cratic adamellites of the Charlotte belt near Salisbury, North Carolina, Geol. Soc. Am. Bull., 89, 460-466, 1978.

Dallmeyer, R. D., J. E. Wright, D. T. Secor, and A. W. Snoke, Character of the Alleghenian orogeny in the southern Appalachians, II, Geochronological constraints on the tectonothermal evolution of the eastern Piedmont in South Carolina, Geol. Soc. Am. Bull., 97, 1329-1344, 1986.

Fisher, R. A., Dispersion on a sphere, Proc. R. Soc. London, Ser. A, 217, 295-305, 1953.

Gibson, G. G., S. A. Teeter, and M. A. Fedonkin, Ediacarian fossils from the Carolina Slate Belt, Stanly County, North Carolina, Geology, 12, 387-390, 1984.

Glover, L. III, J. A. Speer, G. S. Russell, and S. S. Farrar, Ages of regional metamorphism and ductile deformation in the central and southern Appalachians, Lithos, 16, 223-245, 1983.

Hatcher, R. D., Jr., Developmental model for the southern Appalachians, Geol. Soc. Am. Bull., 83, 2735-2760, 1972.

Hatcher, R. D., Jr., and Zietz, I., Tectonic implications of regional aeromagnetic and gravity data from the southern Appalachians, in The Caledonides in the U.S.A., Mem. 2, edited by D. Wones, pp. 235-244, Virginia Polytechnic Institute, Blacksburg, 1980.

Jackson, M. and R. Van der Voo, A lower Ordovician paleomagnetic pole from the Oneota Dolomite, Upper Mississippi River Valley, J. Geophys. Res., 90, 10,449-10,461, 1985.

Kent, D. V. and N. D. Opdyke, Paleomagnetism of Siluro-Devonian rocks from eastern Maine, Can. J. Earth Sci., 17, 1653-1665, 1980.

Kent, D. V. and N. D. Opdyke, Multicomponent magnetizations from the Mississippian Mauch Chunk Formation of the Central Appalachians and their tectonic implications, J. Geophys. Res., 90, 5371-5383, 1985.

Kirschvink, J. L., The least squares line and plane and the analysis of paleomagnetic data, Geophys. J. Roy. Astron. Soc., 62, 699-718, 1980.

Kish, S. A., J. R. Butler, and P. D. Fullager, The timing of metamorphism and deformation in the central and eastern Piedmont of North Carolina, Geol. Soc. Am. Abstr. Programs, 11 184, 1979.

Kligfield, R., W. Lowrie, A. M. Hirt, and A. W. B. Siddans, Effect of progressive deformation on remanent magnetization of Permian redbeds from the Alpes Maritimes (France), Tectonophysics, 97, 59-85, 1983.

Lowrie, W., A. M. Hirt, and R. Kligfield, Effects of tectonic deformation on the remanent magnetization of rocks. Tectonics, 5, 713-722, 1986.

McCabe, C., R. Van der Voo, B. Wilkinson, and K. Devaney, A middle/late Silurian paleomagnetic pole from limestone reefs of the Wabash Formation, Indiana, USA, J. Geophys. Res., 90, (B4), 2959-2965, 1985.

Milton, D. J., Revision of the Albemarle Group in U.S. Geol. Surv. Bull., 1537-A, A69-A72, 1984.

Noel, J. R., D. J. Sparlosu, and R. D. Dallmeyer, Paleomagnetism and $^{40}Ar/^{39}Ar$ ages from the Carolina Slate Belt Albemarle, N.C.: Implications for terrane history, Geology, in press, 1987.

North Carolina Geological Survey, Geologic map of North Carolina, Dep. of Nat. Resour. and Community Dev., Raleigh, 1985.

Perroud, H., R. Van der Voo, and N. Bonhommet, Paleozoic evolution of the Armorica plate on the basis of paleomagnetic data, Geology, 12, 579-582, 1984.

Proko, M. S. and R. B. Hargraves, Paleomagnetism of the Beemerville (New Jersey) complex, Geology, 1, 185-186, 1973.

Randazzo, A. F., Petrography and stratigraphy of the Carolina Slate Belt, Union County, North Carolina, Spec. Publ. 4, N. C. Dep. of Nat. and Econ. Res., Raleigh, 1972.

Rao, K. V. and R. Van der Voo, Paleomagnetism of a Paleozoic anorthosite from the Appalachian Piedmont, northern Delaware: possible tectonic implications, Earth Planet. Sci. Lett., 47, 113-120, 1980.

Secor, D. T., Jr., S. L. Samson. A. W. Snoke, and A. W. Palmer, Confirmation of the Carolina Slate Belt as an exotic terrane, Science, 221, 649-651, 1983.

Secor, D. T., Jr., A. W. Snoke, and R. D. Dallmeyer,

19449194, 1987, 5, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/TC006i005p00573 by Curtin University Library, Wiley Online Library on [25/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Character of the Alleghenian orogeny in the southern Appalachians, III, Regional tectonic relations, Geol. Soc. Am. Bull., 97, 1345-1353, 1986.
Stromquist, A. A., and H. W. Sundelius, Stratigraphy of the Albemarle Group of the Carolina Slate Belt in central North Carolina, U.S. Geol. Surv. Bull., 1274B, 22 pp., 1969.
Sutter, J. F., D. J. Milton, and M. J. Kunk, $^{40}Ar/^{39}Ar$ age spectrum dating of gabbro plutons and surrounding rocks in the Charlotte belt of North Carolina, Geol. Soc. Am. Abstr. Programs, 15, 110, 1983.
Van der Voo, R., and R. B. French, Paleomagnetism of the late Ordovician Juniata Formation and the remagnetization hypothesis, J. Geophys. Res., 82, (36), 5796-5802, 1977.
Van der Voo, R., and R. J. E. Johnson, Paleomagnetism of the Dunn Point Formation (Nova Scotia): High paleolatitudes for the Avalon terrane in the late Ordovician, Geophys. Res. Lett., 12, 337-340, 1985.
Van der Voo, R., R. B. French, and D. W. Williams, Paleomagnetism of the Wilberns Formation (Texas) and the Late Cambrian paleomagnetic field for North America, J. Geophys. Res., 81, 5633-5638, 1976.
Watts, D. R., and R. Van der Voo, Paleomagnetic results from the Ordovician Moccasin, Bays, and Chapman Ridge Formations of the Valley and Ridge Province, eastern Tennessee, J. Geophys. Res., 84, (B2), 645-655, 1979.
Williams, H. and R. D. Hatcher, Jr., Suspect terranes and accretionary history of the southern Appalachian orogen, Geology, 10, 530-536, 1982.
Wright, J. E., and V. M. Seiders, Age of zircons from volcanic rocks of the central North Carolina Piedmont and tectonic implications, Geol. Soc. Am. Bull., Part 1, 91, 287-294, 1980.
Zen, E-an, An alternative model for the development of the allochthonous southern Appalachian Piedmont, Am. J. Sci., 281, 1153-1163, 1981.

J. E. T. Channell, N. D. Opdyke, and H. K. Vick, Department of Geology, University of Florida, Gainesville, FL 32611.

(Received February 12, 1987;
revised May 18, 1987;
accepted May 19, 1987.)

19449194, 1987, 5, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/TC006i005p00573 by Curtin University Library, Wiley Online Library on [25/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License